

B5 Inequalities

Name: _____

Class: _____

Date: _____

Time: **245 minutes**

Marks: **205 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

- (a) Express $2x^2 - 5x + k$ in the form $a(x - b)^2 + c$

(3)

- (b) Find the values of k for which the curve $y = 2x^2 - 5x + k$ does **not** intersect the line $y = 3$

(3)

(Total 6 marks)

Q2.

Consider the two statements, A and B, below.

A: $x^2 - 6x + 8 > 0$

B: $x > 4$

Choose the most appropriate option below.

Circle your answer.

$A \Rightarrow B$

$A \Leftarrow B$

$A \Leftrightarrow B$

There is no
connection
between A and B

(Total 1 mark)

Q3.

Chris claims that, "for any given value of x , the gradient of the curve

$y = 2x^3 + 6x^2 - 12x + 3$ is always greater than the gradient of the curve $y = 1 + 60x - 6x^2$."

Show that Chris is wrong by finding all the values of x for which his claim is **not** true.

(Total 7 marks)

Q4.

The quadratic equation

$$(2k - 7)x^2 - (k - 2)x + (k - 3) = 0$$

has real roots.

- (a) Show that $7k^2 - 48k + 80 \leq 0$.

(4)

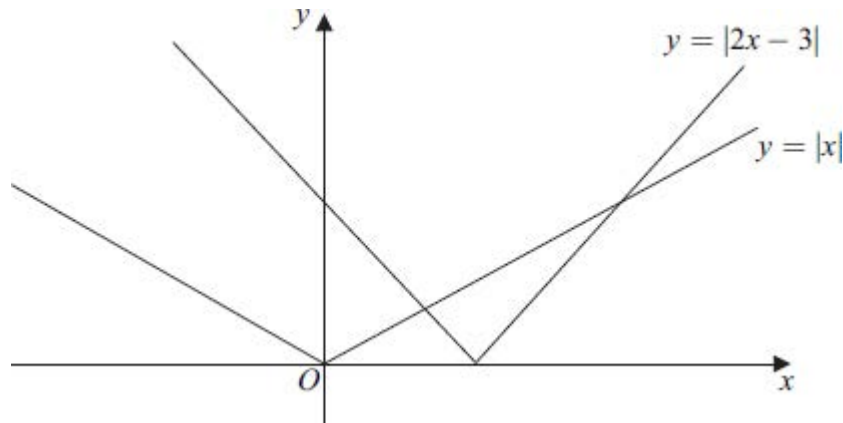
- (b) Find the possible values of k .

(4)

(Total 8 marks)

Q5.

The diagram below shows the graphs of $y = |2x - 3|$ and $y = |x|$.



- (a) Find the x -coordinates of the points of intersection of the graphs of $y = |2x - 3|$ and $y = |x|$.

(3)

- (b) Hence, or otherwise, solve the inequality

$$|2x - 3| \geq |x|$$

(2)

(Total 5 marks)

Q6.

A rectangular garden is to have width x metres and length $(x + 4)$ metres.

- (a) The perimeter of the garden needs to be greater than 30 metres.

Show that $2x > 11$.

(1)

- (b) The area of the garden needs to be less than 96 square metres.

Show that $x^2 + 4x - 96 < 0$.

(1)

- (c) Solve the inequality $x^2 + 4x - 96 < 0$.

(4)

- (d) Hence determine the possible values of the width of the garden.

(1)

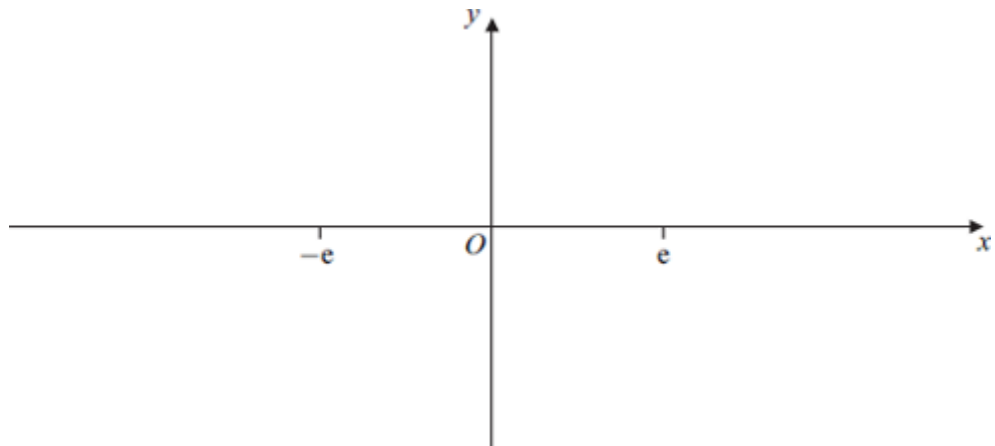
(Total 7 marks)

Q7.

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = \ln x$ onto the graph of $y = 4 \ln(x - e)$.

(4)

- (b) Sketch, on the axes given below, the graph of $y = |4 \ln(x - e)|$, indicating the exact value of the x -coordinate where the curve meets the x -axis.



(3)

- (c) (i) Solve the equation $|4 \ln(x - e)| = 4$.

(3)

- (ii) Hence, or otherwise, solve the inequality $|4 \ln(x - e)| \geq 4$.

(3)

(Total 13 marks)

Q8.

The gradient, $\frac{dy}{dx}$, of a curve C at the point (x, y) is given by

$$\frac{dy}{dx} = 20x - 6x^2 - 16$$

- (a) (i) Show that y is increasing when $3x^2 - 10x + 8 < 0$.

(2)

- (ii) Solve the inequality $3x^2 - 10x + 8 < 0$.

(4)

- (b) The curve C passes through the point $P(2, 3)$.

- (i) Verify that the tangent to the curve at P is parallel to the x -axis.

(2)

- (ii) The point $Q(3, -1)$ also lies on the curve. The normal to the curve at Q and the tangent to the curve at P intersect at the point R . Find the coordinates of R .

(7)

(Total 15 marks)

Q9.

- (a) (i) Express $4 - 10x - x^2$ in the form $p - (x + q)^2$ (2)

- (ii) Hence write down the equation of the line of symmetry of the curve with equation $y = 4 - 10x - x^2$. (1)

- (b) The curve C has equation $y = 4 - 10x - x^2$ and the line L has equation $y = k(4x - 13)$, where k is a constant.

- (i) Show that the x -coordinates of any points of intersection of the curve C with the line L satisfy the equation

$$x^2 + 2(2k + 5)x - (13k + 4) = 0$$

(1)

- (ii) Given that the curve C and the line L intersect in two distinct points, show that

$$4k^2 + 33k + 29 > 0$$

(3)

- (iii) Solve the inequality $4k^2 + 33k + 29 > 0$. (4)

(Total 11 marks)

Q10.

Solve each of the following inequalities:

- (a) $2(4 - 3x) > 5 - 4(x + 2)$; (2)

- (b) $2x^2 + 5x \geq 12$. (4)

(Total 6 marks)

Q11.

A circle with centre C has equation $x^2 + y^2 - 4x + 12y + 15 = 0$.

- (a) Find:
- (i) the coordinates of C ; (2)

- (ii) the radius of the circle. (2)

- (b) Explain why the circle lies entirely below the x -axis. (2)

(c) The point P with coordinates $(5, k)$ lies outside the circle.

(i) Show that $PC^2 = k^2 + 12k + 45$.

(2)

(ii) Hence show that $k^2 + 12k + 20 > 0$.

(1)

(iii) Find the possible values of k .

(4)

(Total 13 marks)

Q12.

(a) Sketch the graph of $y = |8 - 2x|$.

(2)

(b) Solve the equation $|8 - 2x| = 4$.

(2)

(c) Solve the inequality $|8 - 2x| > 4$.

(2)

(Total 6 marks)

Q13.

(a) (i) Express $2x^2 - 20x + 53$ in the form $2(x - p)^2 + q$, where p and q are integers.

(2)

(ii) Use your result from part (a)(i) to explain why the equation $2x^2 - 20x + 53 = 0$ has no real roots.

(2)

(b) The quadratic equation $(2k - 1)x^2 + (k + 1)x + k = 0$ has real roots.

(i) Show that $7k^2 - 6k - 1 \leq 0$.

(4)

(ii) Hence find the possible values of k .

(4)

(Total 12 marks)

Q14.

(a) Factorise $2x^2 - 5x + 3$.

(1)

(b) Hence, or otherwise, solve the inequality $2x^2 - 5x + 3 < 0$.

(3)

(Total 4 marks)

Q15.

The 25th term of an arithmetic series is 38.

The sum of the first 40 terms of the series is 1250.

- (a) Show that the common difference of this series is 1.5.

(6)

- (b) Find the number of terms in the series which are less than 100.

(3)

(Total 9 marks)

Q16.

The curve C has equation $y = k(x^2 + 3)$, where k is a constant.

The line L has equation $y = 2x + 2$.

- (a) Show that the x -coordinates of any points of intersection of the curve C with the line L satisfy the equation

$$kx^2 - 2x + 3k - 2 = 0$$

(1)

- (b) The curve C and the line L intersect in two distinct points.

- (i) Show that

$$3k^2 - 2k - 1 < 0$$

(4)

- (ii) Hence find the possible values of k .

(4)

(Total 9 marks)

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Q17.

The curve C has equation $y = x^2 + 7$. The line L has equation $y = k(3x + 1)$, where k is a constant.

- (a) Show that the x -coordinates of any points of intersection of the line L with the curve C satisfy the equation

$$x^2 - 3kx + 7 - k = 0$$

(1)

- (b) The curve C and the line L intersect in two distinct points. Show that

$$9k^2 + 4k - 28 > 0$$

(3)

- (c) Solve the inequality $9k^2 + 4k - 28 > 0$.

(4)

(Total 8 marks)

Q18.

The quadratic equation $(k + 1)x^2 + 4kx + 9 = 0$ has real roots.

- (a) Show that $4k^2 - 9k - 9 \geq 0$.

(3)

- (b) Hence find the possible values of k .

(4)

(Total 7 marks)

Q19.

The quadratic equation $(k + 1)x^2 + 12x + (k - 4) = 0$ has real roots.

- (a) Show that $k^2 - 3k - 40 \leq 0$.

(3)

- (b) Hence find the possible values of k .

(4)

(Total 7 marks)

Q20.

- (a) Sketch the graph of $y = |2x|$.

(1)

- (b) On a separate diagram, sketch the graph of $y = 4 - |2x|$, indicating the coordinates of the points where the graph crosses the coordinate axes.

(3)

- (c) Solve $4 - |2x| = x$

(3)

- (d) Hence, or otherwise, solve the inequality $4 - |2x| > x$.

(2)

(Total 9 marks)

Q21.

The quadratic equation

$$(2k - 3)x^2 + 2x + (k - 1) = 0$$

where k is a constant, has real roots.

(a) Show that $2k^2 - 5k + 2 \leq 0$.

(3)

(b) (i) Factorise $2k^2 - 5k + 2$.

(1)

(ii) Hence, or otherwise, solve the quadratic inequality

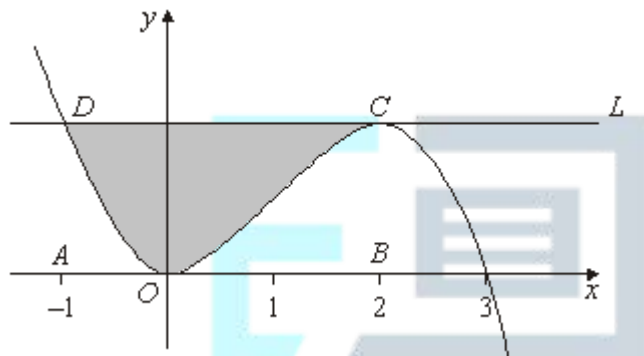
$$2k^2 - 5k + 2 \leq 0.$$

(3)

(Total 7 marks)

Q22.

The diagram shows the curve with equation $y = 3x^2 - x^3$ and the line L .



The points A and B have coordinates $(-1, 0)$ and $(2, 0)$ respectively. The curve touches the x -axis at the origin O and crosses the x -axis at the point $(3, 0)$. The line L cuts the curve at the point D where $x = -1$ and touches the curve at C where $x = 2$.

(a) Find the area of the rectangle $ABCD$.

(2)

(b) (i) Find $\int (3x^2 - x^3) dx$.

(3)

(ii) Hence find the area of the shaded region bounded by the curve and the line L .

(4)

(c) For the curve above with equation $y = 3x^2 - x^3$:

(i) find $\frac{dy}{dx}$;

(2)

(ii) hence find an equation of the tangent at the point on the curve where $x = 1$;

(3)

(iii) show that y is decreasing when $x^2 - 2x > 0$.

(2)

(d) Solve the inequality $x^2 - 2x > 0$.

(2)

(Total 18 marks)

Q23.

(a) Simplify $(k + 5)^2 - 12k(k + 2)$.

(2)

(b) The quadratic equation $3(k + 2)x^2 + (k + 5)x + k = 0$ has real roots.

(i) Show that $(k - 1)(11k + 25) \leq 0$.

(5)

(ii) Hence find the possible values of k .

(3)

(Total 10 marks)

Q24.

Solve each of the following inequalities:

(a) $3(x - 1) > 3 - 5(x + 6)$;

(3)

(b) $x^2 - x - 6 < 0$.

(4)

(Total 7 marks)

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